



Westworld Season 2

Evan Rachel Wood confirmed via Twitter that Westworld's second season is still slated to air in spring 2018. <http://dgit.in/WWSpring18>



Mad Max director suing Warner Bros

Director George Miller is suing Warner Bros. over claims of lost earnings with the highly successful movie <http://dgit.in/MMaxvSWB>

at Tata Elxsi, is the Robot Operating System (ROS). Technically, it is a collection of frameworks for development of robot software, but it also provides services designed for a heterogeneous computer cluster such as hardware abstraction, low-level device control, message-passing between processes, and package management, making it almost a full-fledged operating system.

robot equipped with a pair of wheels, a camera, a laser scanner, and the ROS navigation stack, and you will be learning in a few hours tops.

You can download the ROS framework at ros.org, learn basic and advanced concepts at <http://dgit.in/RIAROS>. Also check out TurtleSim (<http://dgit.in/TurtleSim>) which is a tool made for teaching ROS.

IS IT WORTH IT?

Companies like Hi-Tech Robotics and NXP Technologies are investing time and effort to make this a more mature field in India, and at the same time, bringing the workforce to a standard skill level. "Majority of companies are picking up the pace with autonomous vehicle technology and adopting processes that are aligned to the technological evolution," says Ritukar Vijay, Head of Technology and Strategy, Hi-Tech Robotics. "We at Hi-Tech Robotic Systemz lay huge emphasis on the core knowledge of artificial intelligence and autonomous navigation technology. We aim to provide young aspirants with practical experience in the field of AI and advanced autonomous technology. We happen to work with one of the leaders in self-driving courses online – Udacity."

It's not just road vehicles that can be autonomous. Industrial logistics, navigating hazardous environments and even military vehicles are perfect use cases that are being worked on by startups and big corporations alike, and there's big money and a



Udacity's Self Driving Car Nanodegree is a comprehensive course to prepare yourself for a career in this industry

ROS

ROS: The Linux for Robotics

There are quite a few milestones that ROS has achieved in the autonomous vehicle industry. For instance, Baidu's open sourced autonomous vehicle platform, Apollo, runs entirely on ROS, and BMW, Bosch, and the recent Delphi-acquired NuTonomy also use it.

Advantages of using ROS include:

- **A lot of code is available**
Algorithms with all the capabilities required for the navigation of wheeled robots have already been created in ROS and self-driving cars can just make use of them.
- **Visualization tools present**
A suite of graphical tools for easy recording and visualization of sensor data to represent the status of the vehicle are already available in ROS.
- **It is relatively simple**
You can start right now with a simple

OPPORTUNITIES

You might think India doesn't have a market for self-driving vehicle technologies, and you'd be partially right. After all, the driving conditions in India are far from predictable.

"We are pretty far from making cars fit to drive in everyday Indian traffic conditions," says Mr. Ankur Pandey, "To progress in this industry we need to design increasingly smarter algorithms (AI), but we also need innovation in sensors/hardware components. Then there's another angle, which might be to develop 3D maps with ultra high precision and almost real time updates, which can be used by a vehicle connected to the cloud. It's what we're aiming at achieving at HERE Technologies."

Autonomous vehicles actually perform better when there are more autonomous vehicles in operation. They're able to communicate and learn from one another. Of course, we need the required infrastructure and policies, which is something that the Government of India is still working on.



CSIRO ICT Centre has trialled its automation technology in large Hot Metal Carriers (HMC), capable of delivering buckets of molten aluminium without human intervention, at Rio Tinto Aluminium's Bell Bay Smelter in Tasmania.

lucrative career in those fields. Now, it is up to you whether you want to board the driverless bus to the future, or stick with tried and tested fields. Either way, here's wishing you a safe (and hands-free) journey.